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MIL-H-19387A(SHIPS)
20 July 1970
SUPERSEDING
MIL-H-19387(SHIPS)
9 October 1956
(See 6.4)

MILITARY SPECIFICATION

HEADS, SPRINKLER AND WASHDOWN

1. SCOPE

1.1 Scope. This specification covers sprinkler heads for use in fire extinguisher sprinkler and decontamination shower systems.

1.2 Classification. Sprinkler heads shall be of the following types and classes, as specified (see 6.1):

Type I - For pendant on upright installation in overhead sprinkler system (see figure 1).
Type II - Watercurtain or barrier type for installation in fixed piping system.
Class 1 - Intermediate velocity (see figures 2 and 3).
Class 2 - High velocity (see figures 4 and 5).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
QQ-A-591 - Aluminum Alloy Special Shaped Section.
QQ-C-390 - Copper Alloy Castings (Including Cast Bar).
PPP-B-636 - Boxes, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.

MILITARY
MIL-P-116 - Preservation, Methods of.
MIL-F-1183 - Fittings, Tube; Cast Bronze Silver Brazing.

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

NATIONAL CLASSIFICATION BOARD
National Motor Freight Classification Rules

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., 1616 P Street, N. W., Washington, D. C. 20036.)

FSC 4510

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3. REQUIREMENTS

3.1 Sprinkler heads shall meet the delivery rates and patterns shown on figures 1 through 5.

3.2 Material. Sprinkler head bodies shall be made of copper alloy conforming to QQQ-C-390, alloy G7, or aluminum alloy conforming to QQ-A-591, as specified (see 6.1). Deflector plates and supporting arms shall be of copper alloy for bronze heads and aluminum alloy for aluminum heads.

3.3 Construction.

3.3.1 Type I sprinkler heads (see figure 1). Type I sprinkler heads shall have a single open orifice directing the stream to impinge upon an external deflector plate firmly supported and protected by two cross yokes rigidly fixed to or integral with the head body. The deflector plate shall be suitably designed to produce a homogeneous conical spray. The body shall be designed with a 1/2-inch iron pipe size (i.p.s.) male thread for installation in plastic or aluminum fittings or 1/2 inch shank for installation with silver brazed fittings in accordance with MIL-F-1183 as specified (see 6.1). The size of the orifices and angle of the cone shall be as specified (see 6.1). The capacities of the heads shall meet the flow characteristics specified in table I.

Table I - Discharge capacity for type I sprinkler heads.

Orifice size	Capacity gallons per minute (g.p.m.)			
	50 pounds per square inch (p.s.i.)		100 pounds per square inch (p.s.i.)	
	Minimum	Maximum	Minimum	Maximum
16	7	9	11	13
18	13	15	18	20
21	15	17	21	25
24	20	24	28	32
28	26	30	37	41
32	36	40	52	58

3.3.2 Type II sprinkler heads.

3.3.2.1 Class 1 (see figure 2). Class 1 intermediate velocity sprinkler heads shall be so designed that the single open orifice directs the stream to impinge upon a deflector plate integral with 90 degrees to the body of the head to produce a water curtain effect. The body shall have a 1/2-inch i.p.s. male thread or 0.840 inch outside diameter (o.d.) shank as specified (see 6.1) for installation in piping system. The size of the orifice and spread of the water curtain shall be as specified (see 6.1). Figure 3 indicates typical pattern and flow characteristics of intermediate velocity sprinkler heads.

3.3.2.2 Class 2 (see figure 4). Class 2 high velocity sprinkler heads shall be so designed that the single open orifice directs the stream to impinge upon a deflector plate integral with and at a suitable angle to the body of the head to produce a fan-shaped water curtain pattern. The body shall have a 1/2-inch i.p.s. male thread or 0.840 inch o.d. shank as specified (see 6.1) for installation in the piping system. The size of the orifice and spread of the water curtain shall be as specified (see 6.1). Figure 5 indicates a typical pattern and flow characteristics for high velocity sprinkler heads.

3.4 Workmanship.

3.4.1 All fittings shall be sound, smoothly cored, true to form, uniform in texture, and free from cold shunts, porosity, or any other defects which may affect their serviceability. They shall be thoroughly cleaned, both inside and outside, and all fins and roughness shall be removed.

3.4.2 Castings shall not be repaired, plugged, or welded without permission from the Government representative, which shall be given only when the defects are small (less than 10 percent of a size which would cause the fittings to be rejected) and do not adversely affect the strength, use, or machinability of the castings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the

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supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lot. All sprinkler heads of the same type, class, and orifice size offered for delivery at one time shall be considered a lot for purposes of inspection.

4.2.2 Sampling for visual and dimensional examination. A random sample of sprinkler heads shall be selected from each lot in accordance with MIL-STD-105 at inspection level II for the examination of 4.3. The Acceptable Quality Level shall be 1.5 percent defective. Any sprinkler head in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective sprinkler heads in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.2.3 Sampling for tests. A random sample of sprinkler heads shall be selected from each lot in accordance with MIL-STD-105 at inspection level III for lots of 65 and under, inspection level II for lots of 66 to 500, and inspection level I for lots of 501 and over for the tests of 4.4. The Acceptable Quality Level shall be equal to 2.5 percent defective. Any sample not meeting the requirements of this specification shall be rejected, and if the number of nonconforming sprinkler heads in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.3 Visual and dimensional examination. Each of the sample sprinkler heads selected in accordance with 4.2.2 shall be visually and dimensionally examined to verify compliance with this specification.

4.4 Test procedures.

4.4.1 Each type I sample sprinkler head shall be subjected to a performance test to determine compliance with the flow characteristics specified herein.

4.4.2 Each type II sample sprinkler head shall be subjected to a performance test to determine compliance with the flow and pattern characteristics specified herein.

4.5 Inspection of preparation for delivery. The packaging, packing, and marking shall be inspected for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.3.)

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.1.1 Level A. The sprinkler heads shall be individually packaged in a folding, set-up paperboard or corrugated fiberboard box in accordance with method III of MIL-P-116.

5.1.2 Level C. The sprinkler heads shall be individually packaged to afford protection against deterioration and physical damage during shipment from the supply source to the first receiving activity for immediate use. The supplier's normal preservation-packaging methods may be utilized when such meet the requirements of this level.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.2.1 Level A. Sprinkler heads packaged as specified shall be packed in a fiberboard box conforming to PPP-B-636 class-weather resistant, grade V, or a triple wall corrugated fiberboard box conforming to PPP-B-640, class 2. Variety and style of the box shall be at the supplier's option. Boxes shall be waterproofed in accordance with the appendix to the box specification. Adhesive shall not be used for box closure. Boxes shall be reinforced with nonmetallic or tape banding. The gross weight of fiberboard boxes shall not exceed the weight limitation of the applicable box specification.

5.2.2 Level B. Sprinkler heads packaged as specified shall be packed in a fiberboard box conforming to PPP-B-636 class-domestic, or a triple wall corrugated box conforming to PPP-B-640, class 1. Variety and style of the box shall be at the supplier's option. Adhesive shall not be used for box closure. The gross weight of fiberboard boxes shall not exceed the weight limitation of the box specification.

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5.2.3 Level C. Sprinkler heads packaged as specified (see 6.1), shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest applicable rate. Containers, packing or method of shipment shall comply with Uniform Freight and National Motor Freight Classification Rules and Regulations or other carrier rules as applicable to the mode of transportation.

5.3 Marking. In addition to any special marking specified in the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.4 Use of polystyrene (loose-fill) material.

5.4.1 For domestic shipment and early equipment installation and level C packaging and packing. Unless otherwise approved by the procuring activity (see 6.1), use of polystyrene (loose-fill) material for domestic shipment and early equipment installation and level C packaging and packing applications such as cushioning, filler and dunnage is prohibited. When approved, unit packages and containers (interior and exterior) shall be marked and labelled as follows:

"CAUTION

Contents cushioned etc with polystyrene (loose-fill) material.
Not to be taken aboard ship.
Remove and discard loose-fill material before shipboard storage.
If required, recushion with cellulosic material bound fiber,
fiberboard or transparent flexible cellular material."

5.4.2 For level A packaging and level A and B packing. Use of polystyrene (loose-fill) material is prohibited for level A packaging and level A and B packing applications such as cushioning, filler and dunnage.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and class required (see 1.2).
(c) Whether aluminum alloy or copper alloy material is required (see 3.2).
(d) Whether sprinkler head shall have 1/2 inch male thread or a shank for installation in fitting (see 3.3.1, 3.3.2.1, and 3.3.2.2).
(e) Diameter of orifice and angle of the deflector (see 3.3.1, 3.3.2.1 and 3.3.2.2).
(f) Spread of the water curtain (see 3.3.2.1 and 3.3.2.2).
(g) Level of packaging and packing required (see 5.1 and 5.2).
(h) When polystyrene (loose-fill) material is approved (see 5.4.1).

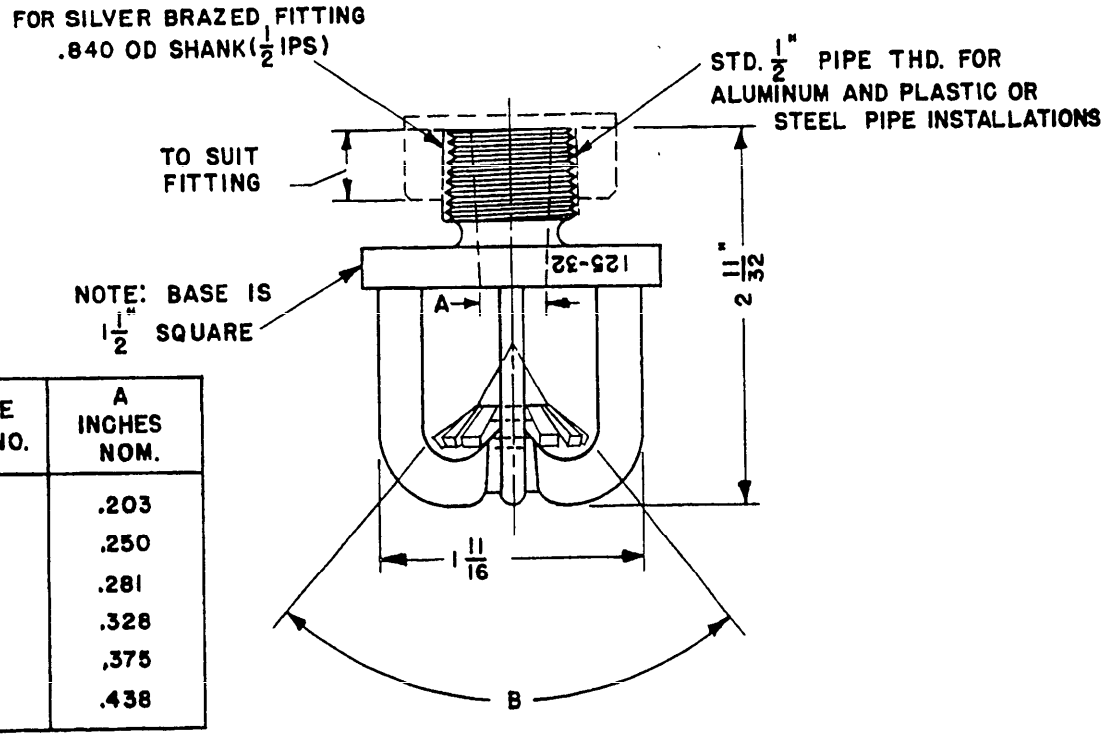
6.2 The installation of silver brazed fittings, is intended to be accomplished by silver brazing without exceeding a temperature of 1400°F. The fittings are intended for use with copper-nickel tubing of MIL-T-16420.

6.3 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.4 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

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ORIFICE SIZE NO.	A INCHES NOM.
16	.203
18	.250
21	.281
24	.328
28	.375
32	.438

'A' = ORIFICE SIZE. NOS. 16, 18, 21, 24, 28, & 32.

'B' = INCLUDED ANGLE OF DISCHARGE. 160°, 140°, 125°, 110°, 95°, & 80°. (DEFLECTOR PLATE ANGLE IS GREATER THAN DISCHARGE ANGLE)

ANGLE OF DISCHARGE AND ORIFICE SIZE STAMPED ON ONE SIDE OF BASE THUS: 125-32

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Figure 1 - Type I sprinkler head.

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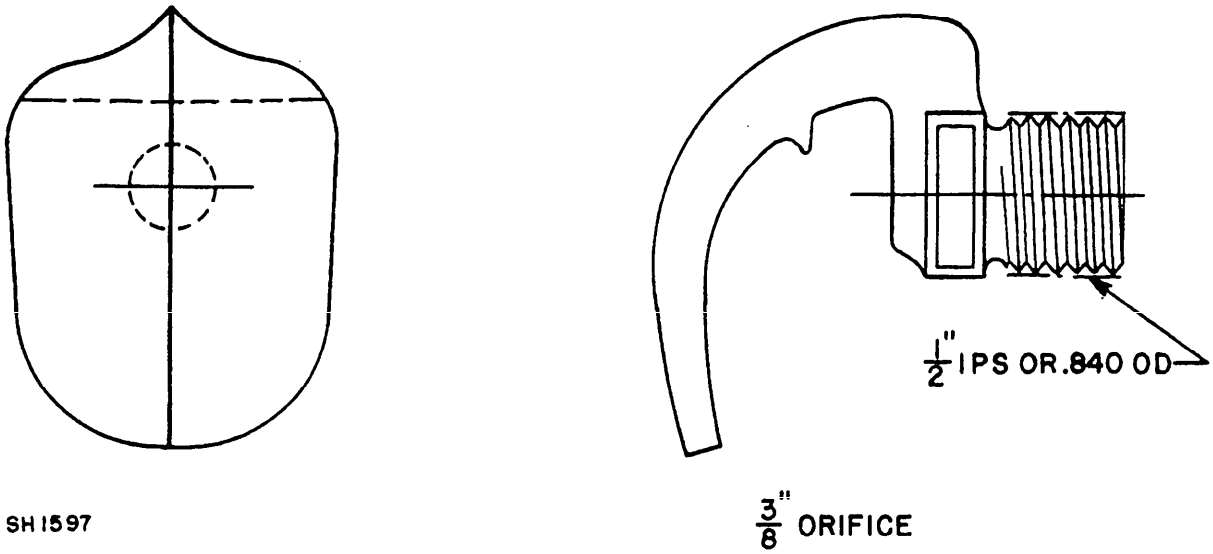


Figure 2 - Type II, class 1, intermediate velocity water curtain sprinkler head.

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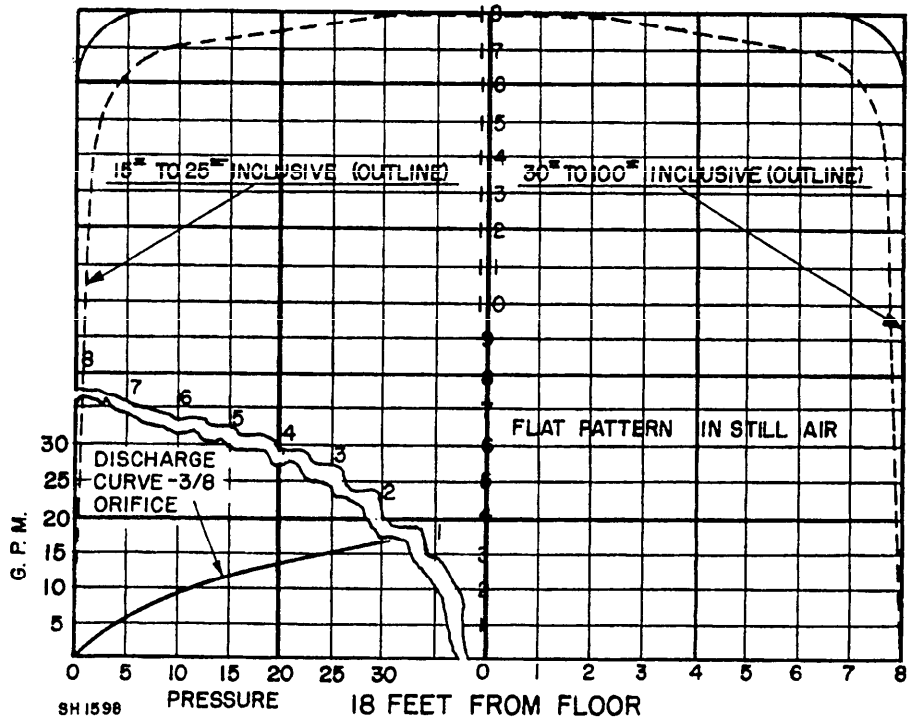


Figure 3 - Water spray pattern for type II, class 1, intermediate water curtain sprinkler head.

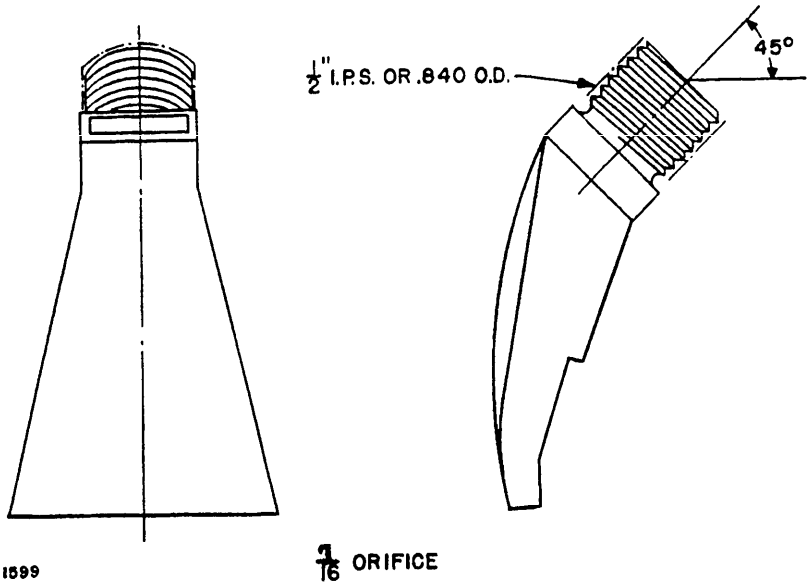


Figure 4 - Type II, class 2, high velocity water curtain sprinkler head.

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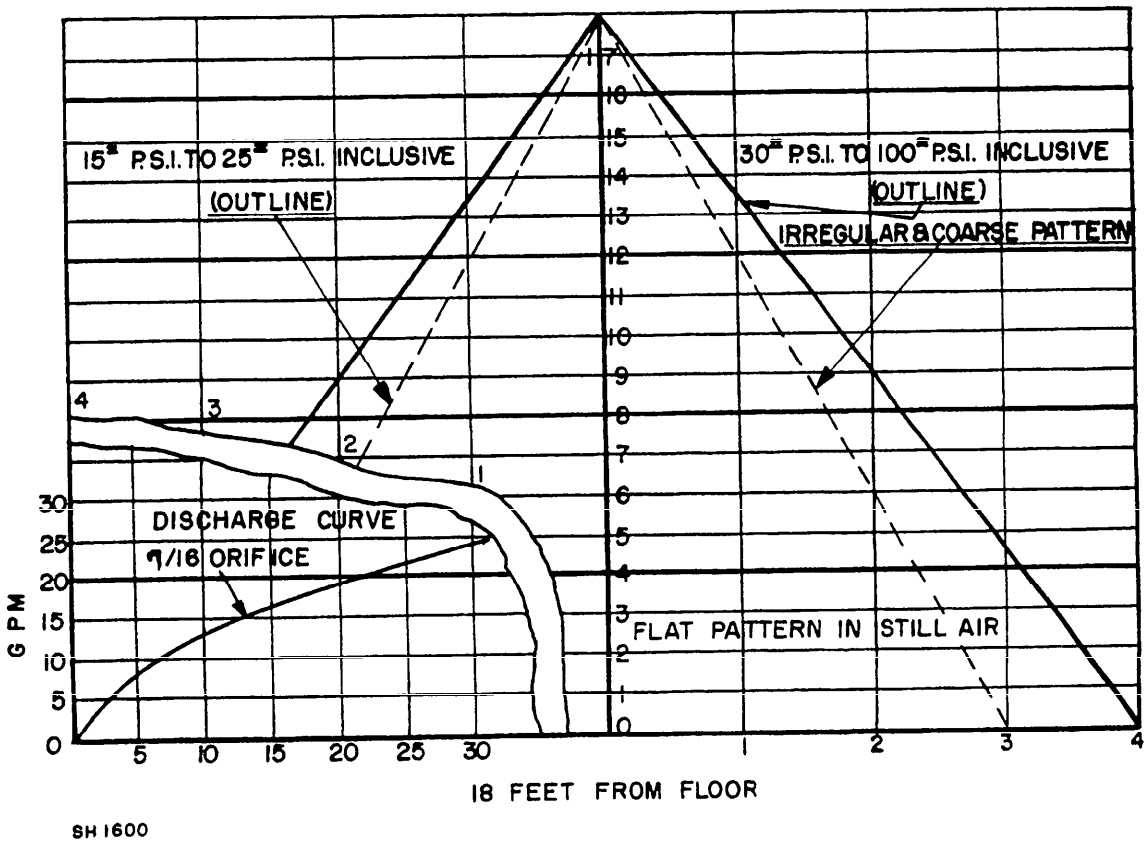


Figure 5 - Water spray pattern for type II, class 2, high velocity water curtain sprinkler head.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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